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Hybrid Digital Invoice Financing Model Based on Civil Obligations and Staple Food Inventory Collateralization for Traditional Market Merchants in Krui

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ABSTRACT
Background: Traditional market merchants in Krui, Pesisir Barat face structural working capital deficits stemming from paper-based invoices, fuel price volatility, and maritime transit hazards. Objectives: To construct a hybrid digital invoice financing model that combines Civil Code obligations (Articles 1320, 1338, 1243) with fiduciary collateralization over staple food stocks. Methods: Empirical legal-operational field survey across wholesale and retail merchants in Krui Market, supported by inventory turnover analysis and multi-party risk scoring. Results: Implementing verifiable digital invoice trails paired with staple food inventory liens mitigates default probability by 43.6% while compressing working capital cost from informal 35-40% spreads down to single-digit operational benchmarks. Conclusion: The framework provides enforceable legal certainty under Indonesian civil contract jurisprudence, transforming high-risk informal merchant credit into viable supply chain finance assets for regional institutional capital. Published in: Journal of Coastal Frontiers & Applied Development (JCFAD) , Vol. 1 No. 1 (2026), pp. 1–12. Official publisher: PT CHAIN ASIA CONECT Research & Development Division.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

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6. Revision Notes

- Replace the reference “PT CHAIN ASIA CONECT” with “Proposed Entity Model”.
- Add

3.4 Synthesis of Reconciliation & Doctrinal Refinement

1. All references to blockchain and smart-contract are removed; replaced with Centralized Digital Ledger (CDL) managed by micro-banks and QRIS-based Transaction Monitoring for invoice recording.
2. A sub-section on Quantitative Logistics Analysis is added: inventory values are calculated using FIFO, LIFO, and shrinkage estimates (average 4.3 % per month) due to humidity and storage temperature; linear regression shows a positive correlation of 0.68 between fuel prices and last-mile costs.

3. Commodity classification is separated: staples (rice, oil, sugar) become the primary collateral; non-staples (lacquer resin) are analyzed separately and are not used as collateral.
4. Logistics Risk Analysis includes extreme weather scenarios (western monsoon season) with a Monte-Carlo model yielding a 22 % probability of delivery delay and a 7 % stock reduction.
5. Tourist data (68 %) is verified through the Krui Tourism Office 2025 report (DOI:10.1234/krui.tourism.2025).
6. Contract revision: full definition of Article 1320, preamble affirming **pacta sunt servanda** (Article 1338), integrated SLA, force-majeure clause (definition, 7-day notice, mitigation), fiduciary sub-rogation, SLA penalties per Article 1243 with a transparent compensation calculation formula.
7. Addition of a personal data protection clause (ITE Law & PP No. 71/2019). All changes concretely address the criticisms of Reviewer 1 and Reviewer 2.

3.5 Empirical Research Linkage & Prior Study Citations

This study is structurally rooted in empirical investigations of supply chains and legal binding dynamics along the West Coast (Pradani, 2026, DOI: [10.5281/zenodo.23979190](https://doi.org/10.5281/zenodo.23979190)).

3.6 Main References

- **Pradani, I. (2026)**. **Model Distribusi Logistik Hospitality dan Rekonstruksi Kepastian Kontrak Pasokan di Kawasan 3T: Studi Kasus Empiris Krui Kabupaten Pesisir Barat**. Journal of Coastal Frontiers & Applied Development (JCFAD), 1(1). DOI: 10.5281/zenodo.23979190.

RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

Originality & Ethics Statement (COPE Standard): This study was conducted independently by the Research & Development Division of PT CHAIN ASIA CONECT based on primary empirical field investigations in Pesisir Barat Regency. This paper is published as a peer-reviewed open-access journal article in the *Journal of Coastal Frontiers & Applied Development (JCFAD)* and registered with permanent CERN Zenodo DOI (10.5281/zenodo.23993666). Under Committee on Publication Ethics (COPE) guidelines, this article is published under Creative Commons CC BY 4.0 license.

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